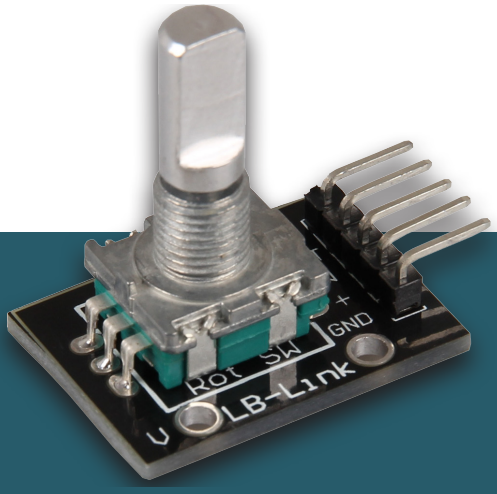


ROTARY ENCODER

COM-KY040RE

Joy-IT powered by SIMAC Electronics GmbH - Pascalstr. 8 - 47506 Neukirchen-Vluyn - www.joy-it.net

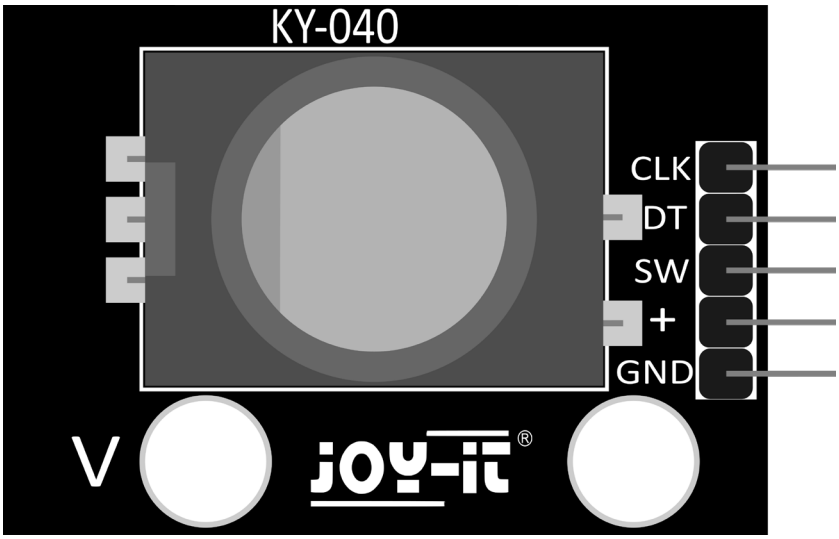
1. GENERAL INFORMATION

Dear customer,
thank you for choosing our product. Below we will show you what you need to bear in mind during commissioning and use.

Should you encounter any unexpected problems during use, please do not hesitate to contact us.

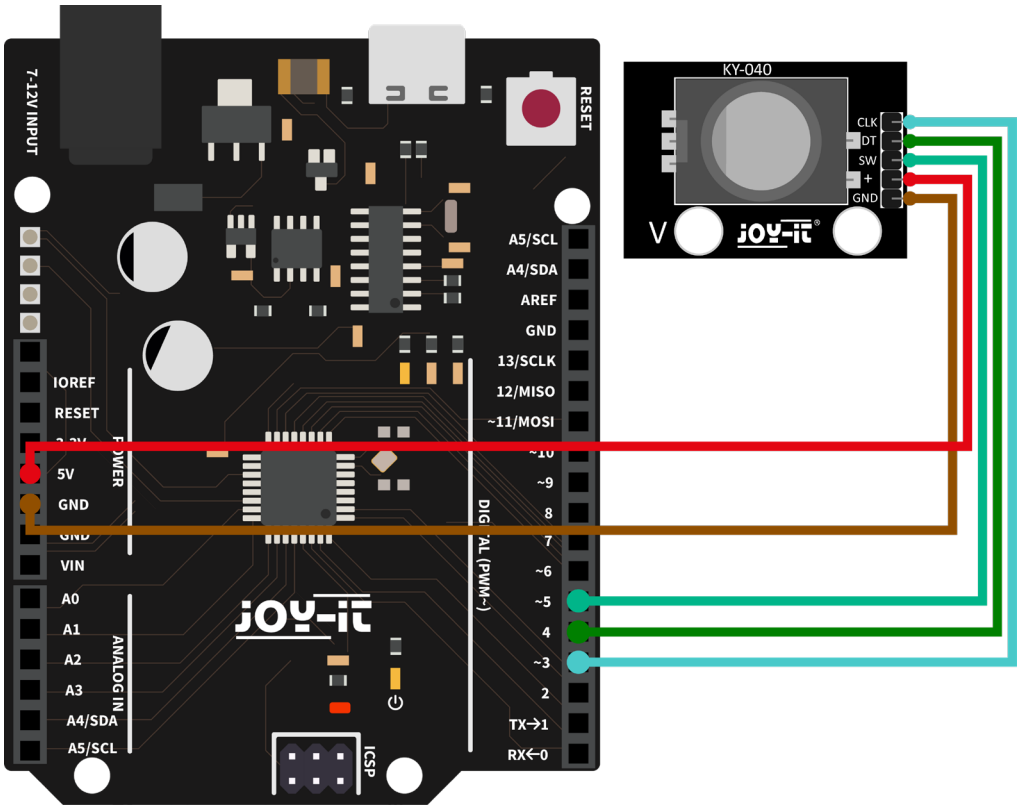
2. EQUIPMENT OVERVIEW

This joystick module outputs the current positions of the X and Y axes as an analog signal via the VRx and VRy pins. The joystick can also be pressed down, which changes the status at pin SW from LOW to HIGH.



3. ARDUINO APPLICATION EXAMPLE

First connect the module to your Arduino:



ROTARY ENCODER	ARDUINO
GND	GND
+	5V
SW	D5
DT	D4
CLK	D3

Now transfer the code example on this and the next page to your Arduino.

```
// Initialization of needed variables
int Counter =  0;
boolean Direction;
int Pin_clk_Last;
int Pin_clk_Current;

// Definition of input pins
int pin_clk =  3;
int pin_dt =  4;
int button_pin =  5;

void setup() {
  // Initialize input pins...
  pinMode(pin_clk, INPUT);
  pinMode(pin_dt, INPUT);
  pinMode(button_pin, INPUT);

  // ...and activate their pull-up resistors
  digitalWrite(pin_clk, true);
  digitalWrite(pin_dt, true);
  digitalWrite(button_pin, true);

  // Initial reading of Pin_CLK
  Pin_clk_Last = digitalRead(pin_clk);
  Serial.begin(115200);
}

/*
The program checks if there has been a change in the pin states, which of
the two
pins changed first, indicating the direction of rotation.
This information is obtained by comparing one of the pin values from a
previous
run with the value of the current run.
Once the direction is determined, steps are counted from the start
position and displayed.
Pressing the button on the rotary encoder resets the current position.
*/
```

```
void loop() {
  // Read the current status
  Pin_clk_Current = digitalRead(pin_clk);

  // Check for change
 if (Pin_clk_Current != Pin_clk_Last) {

   if (digitalRead(pin_dt) != Pin_clk_Current) {
      // Pin_CLK changed first
      Counter++;
      Direction = true;
    }

   else {  // Otherwise, Pin_DT changed first
      Direction = false;
      Counter--;
    }
    Serial.println("Rotation detected: ");
    Serial.print("Rotation direction: ");

   if (Direction) {
      Serial.println("Clockwise");
    } else {
      Serial.println("Counterclockwise");
    }

    Serial.print("Current position: ");
    Serial.println(Counter);
    Serial.println("------------------------------");
  }

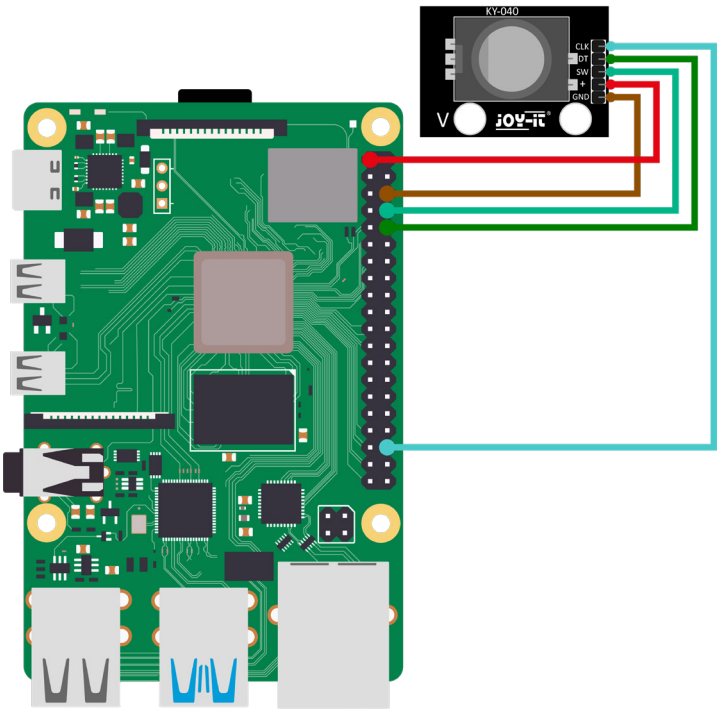
  /*
  Preparation for the next run:
  The value of the current run is the previous value in the next run
  */
  Pin_clk_Last = Pin_clk_Current;

  // Reset function to save the current position
 if (!digitalRead(button_pin) && Counter != 0) {
    Counter = 0;
    Serial.println("Position reset");
  }
}
```

4. APPLICATION EXAMPLE RASPBERRY PI

These instructions were written under Raspberry Pi OS Bookworm for the Raspberry Pi 4 and 5. It has not been checked with newer operating systems or hardware.

First connect the module to your Raspberry Pi.



ROTARY ENCODER	RASPBERRY PI
GND	GND
+	3V3
SW	GPIO14
DT	GPIO15
CLK	GPIO16

You can download the code example that we have made available to you here or you can execute the following command in your console.

```
wget https://www.joy-it.net/files/files/Produkte/COM-KY040RE/COM-KY040RE-RPi.zip
```

Now unpack the file with the following command:

```
unzip COM-KY040RE-RPi.zip
```

You can now execute the script with the following command. Pay attention Please note that your file path may differ.

```
python3 COM-KY040RE-RPi.py
```

Alternatively, you can copy the code example here and paste it into Insert your Python file.

```
from gpiozero import Button
from signal import pause

# Declaration of input pins where the sensor is connected.
PIN_CLK = 16
PIN_DT = 15
BUTTON_PIN = 14

# Creating button objects
button_clk = Button(PIN_CLK)
button_dt = Button(PIN_DT)
button_reset = Button(BUTTON_PIN)

# Initializing necessary variables
Counter = 0
Direction = True
PIN_CLK_LAST = 0

# Initial read of the PIN_CLK
PIN_CLK_LAST = button_clk.value

# This output function executes upon signal detection
def outputFunction():
   global Counter, Direction, PIN_CLK_LAST

    PIN_CLK_CURRENT = button_clk.value

   if PIN_CLK_CURRENT != PIN_CLK_LAST:

       if button_dt.value != PIN_CLK_CURRENT:
            Counter += 1
            Direction = True
       else:
            Direction = False
            Counter -= 1

       print("Rotation detected: ")

       if direction:
           print("Rotation direction: Clockwise")
       else:
           print("Rotation direction: Counterclockwise")

       print("Current position: ", Counter)
       print("------------------------------")
        PIN_CLK_LAST = PIN_CLK_CURRENT

def counter_reset():
   global Counter
   print("Position reset!")
   print("------------------------------")
    Counter = 0

# Add callbacks for the buttons
button_clk.when_pressed = outputFunction
button_dt.when_pressed = outputFunction
button_reset.when_pressed = counter_reset

print("Sensor test [press CTRL+C to end the test]")

# Keep the program running in an infinite loop to process events
pause()
```

5. INFORMATION & TAKE-BACK OBLIGATIONS

Our information and take-back obligations under the German Electrical and Electronic Equipment Act (ElektroG)



Symbol on electrical and electronic equipment:
This crossed-out garbage can means that electrical and electronic appliances **do not** belong in household waste. You must hand in the old appliances at a collection point. Before handing them in, you must separate used batteries and accumulators that are not enclosed by the old appliance.

Return options:
As an end user, you can hand in your old appliance (which essentially fulfills the same function as the new appliance purchased from us) for disposal free of charge when purchasing a new appliance. Small appliances with no external dimensions greater than 25 cm can be disposed of in normal household quantities regardless of whether you have purchased a new appliance.

Possibility of return at our company location during opening hours:
SIMAC Electronics GmbH, Pascalstr. 8, D-47506 Neukirchen-Vluyn

Return option in your area:
We will send you a parcel stamp with which you can return the device to us free of charge. To do so, please contact us by e-mail at Service@joy-it.net or by telephone.

Packaging information:
Please pack your old appliance securely for transportation. If you do not have suitable packaging material or do not wish to use your own, please contact us and we will send you suitable packaging.

6. SUPPORT

We are also there for you after your purchase. If you still have any questions or problems arise, we are also available by e-mail, telephone and ticket support system.

E-Mail: service@joy-it.net
Ticket-System: https://support.joy-it.net
Phone: +49 (0)2845 9360 - 50 (Mon - Thur: 09:00 - 17:00 o'clock CET,
Fri: 09:00 - 14:30 o'clock CET)

For further information, please visit our website:
www.joy-it.net

Published: 2024.05.07